



DEEP LEARNING-BASED
CLASSIFICATION OF PARKINSON'S DISEASE
FROM DATSCAN IMAGES

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BACHELOR'S DEGREE IN BIOMEDICAL ENGINEERING
JUNE 2026

Introduction

SECTION 1

Parkinson's Disease (PD) is the world's fastest-growing neurological disorder and the second most common neurodegenerative disease globally, affecting an estimated 10 million people and projected to nearly double in prevalence over the next two decades as populations age (1,2). The disease also imposes an enormous economic burden: in the United States alone, annual costs, including direct medical expenditure, caregiving, and lost productivity, reached \$82.2 billion in 2024, surpassing projections that had not been expected until 2037 (3).

This diagnostic barrier stems directly from the pathophysiology of the disease. PD is characterized by the progressive degeneration of dopaminergic neurons in the *substantia nigra pars compacta* (SNpc), whose axons project to the striatum through the nigrostriatal pathway to regulate voluntary movement. This degeneration is slow and irreversible, meaning that by the time the motor symptoms, such as resting tremor or rigidity, become clinically apparent, substantial damage has already been done.

Dopamine transporter (DaT) imaging with ^{123}I -ioflupane SPECT, commercially known as DaTscan, has become a key diagnostic tool in this effort. By radiolabeling the dopamine transporter protein on presynaptic terminals, DaTscan provides a direct, functional map of the integrity of the nigrostriatal pathway in vivo. In PD patients, the image generated visualizes degeneration of the nigrostriatal pathway (see FIGURE 2) (4).

Despite its value, DaTscan interpretation in clinical practice relies predominantly on visual assessment by trained specialists, a process that is time-consuming and inherently subjective. Inter-observer discrepancies are well documented, particularly in borderline cases, where early-stage putaminal thinning produces only subtle deviations from normal patterns (5).

Semi-quantitative tools can provide a more objective summary of tracer binding. However, they compress the spatial information of the 3D scan into a small number of regional averages, potentially discarding diagnostically relevant patterns that a more expressive model could exploit (5,6).

The rapid maturation of deep learning offers a compelling path toward addressing these limitations. Convolutional neural networks (CNNs), trained end-to-end on large labeled image datasets, can learn to detect subtle spatial patterns in medical images without relying on manually engineered features, and have demonstrated expert-level performance across a wide range of diagnostic imaging tasks (7). Applied to DaTscan classification, CNN-based approaches may provide more consistent and quantitative diagnostic support, particularly in the early and prodromal stages where visual assessment is least reliable.

This thesis investigates deep learning-based binary classification of manifest PD versus healthy controls utilizing DaTscan images. The data are drawn from the Parkinson's Progression Markers Initiative (PPMI) dataset, which represents one of the largest cohorts available (8).

Preliminary Concepts

SECTION 2

2.1. Parkinson's Disease

2.1.1. Overview and Epidemiology

While the global scaling of Parkinson's disease presents a clear macroeconomic crisis, its clinical footprint is highly dependent on demographic vectors. The disorder exhibits a stark age-dependent incidence profile, with a median age of onset centered around 60 years, though young-onset cases present distinct clinical challenges. Furthermore, epidemiological data consistently reveal a sex-based divergence, with men affected at an approximate ratio of 1.5:1 compared to women (1).

The structural burden of PD is characterized not only by healthcare-related expenditures but also by the hidden economic toll of informal caregiving and premature labor force departure. Because the clinical diagnosis is preceded by an extended prodromal phase, patients frequently navigate a costly diagnostic journey before arriving at a confirmed classification (3). Developing highly objective, automated screening tools is therefore essential to alleviate both the clinical bottlenecks and the compounding economic strains on healthcare infrastructure.

2.1.2. Pathophysiology: The Nigrostriatal Dopaminergic System

The central pathological event in PD is the progressive degeneration of dopaminergic neurons in the *substantia nigra pars compacta* (SNpc), a pigmented region of the midbrain named for the dark neuromelanin granules visible on gross anatomy (see FIGURE 1). These neurons form the nigrostriatal pathway, projecting their axons to terminate in the striatum, a subcortical structure composed of the caudate nucleus and the putamen, where they release dopamine as a neuromodulatory signal within the basal ganglia motor circuit (9).

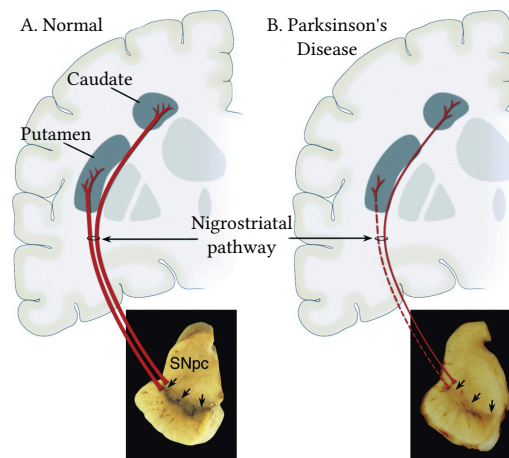


FIGURE 1: Coronal schematic comparison of the nigrostriatal pathway under normal conditions and in Parkinson's disease, illustrating the preferential loss of dopaminergic projections to the putamen and caudate nucleus (dashed lines). Adapted from (9) (pg. 891).

Within the basal ganglia, dopamine modulates two antagonistic pathways: a “direct” or “go” pathway that facilitates movement by disinhibiting the thalamus, and an “indirect” or “stop” pathway that suppresses competing motor programs. Adequate dopamine levels maintain a functional balance between these pathways, enabling smooth, voluntary motion. As SNpc neurons progressively degenerate and striatal dopamine falls, this balance shifts toward inhibition, resulting in the motor deficits characteristic of PD (9).

A clinically critical feature of this process is that motor symptoms do not show until the neurodegenerative process is already advanced. This means there is a substantial presymptomatic period during which pathology is actively progressing but patients are asymptomatic and diagnosis is not possible with current clinical criteria alone.

At the cellular level, the hallmark histopathological finding is the accumulation of Lewy bodies: intraneuronal protein aggregates composed primarily of misfolded alpha-synuclein (aSyn).

2.1.3. Clinical Presentation and Diagnosis

The cardinal motor features of PD: bradykinesia (obligatory for diagnosis under MDS criteria), resting tremor, rigidity, and postural instability, form the basis of clinical evaluation (10). In early disease, motor signs are typically asymmetric, reflecting the asymmetric onset of nigrostriatal degeneration, and a clear positive response to levodopa replacement therapy provides additional diagnostic support.

PD also produces a range of non-motor symptoms that frequently predate the motor phase by years. Hyposmia (reduced olfactory sensitivity, assessed in this thesis using the UPSIT score), REM sleep behavior disorder, constipation, and affective symptoms such as depression and anxiety belong to this prodromal syndrome and are increasingly used in research criteria for preclinical PD (10). The presence of these markers in subjects without motor symptoms is of direct relevance to the PPMI cohort structure and to the multimodal features used in this thesis (see *próximamente en cines*).

The primary diagnostic challenge is the clinical overlap with conditions that mimic parkinsonism: essential tremor (ET), drug-induced parkinsonism (DIP), and atypical parkinsonian syndromes such as multiple system atrophy (MSA), progressive supranuclear palsy (PSP), and corticobasal syndrome (CBS). Consequently, the overall clinical accuracy of a purely symptom-based PD diagnosis remains vulnerable to initial misinterpretation, especially during early presentation. To resolve this diagnostic ambiguity, dopamine transporter (DaTscan) imaging has become a vital objective adjunct to confirm or rule out an underlying neurodegenerative syndrome associated with dopamine deficiency (NSDD).

The real-world impact of this modality on clinical management was demonstrated by Isaacson et al., who examined a cohort of 201 consecutive patients presenting with clinically questionable NSDD. The investigators identified that DaTscan imaging provides crucial clarity across an array of ambiguous clinical scenarios, including patients with subtle early symptoms, prominent action tremors, suspected DIP, or a suboptimal response to levodopa. Notably, imaging also unmasked misdiagnoses in individuals who had carried a PD label for three to five years but failed to show the expected clinical progression or motor fluctuations. Within this uncertain cohort, DaTscan categorized 58.7% of the cases as abnormal, 37.8% as normal, and only 3.5% as inconclusive. Crucially, these objective findings reoriented clinical directions, altering the definitive diagnosis in 39.8% of patients and directly driving medication therapy modifications in 70.1% of cases (11).

2.2. *DaTscan: Imaging the Dopamine Transporter* ---

2.2.1. Mechanism

The dopamine transporter (DAT) is a membrane-bound monoamine transporter expressed exclusively on the terminals of presynaptic dopaminergic neurons. Its physiological role is to recycle released dopamine back into the neuron; its density in the striatum therefore directly reflects the density of viable nigrostriatal terminals and, by extension, the degree to which the pathway has been preserved or destroyed (4).

DaTscan uses ^{123}I -ioflupane (^{123}I -FP-CIT), a radiolabeled cocaine analogue with high affinity for the DAT. Following intravenous injection, the tracer crosses the blood-brain barrier and binds preferentially to DAT in the striatum. Single Photon Emission Computed Tomography (SPECT) acquisition, typically performed 3–6 hours post-injection to allow clearance of non-specific binding, reconstructs a three-dimensional map of tracer uptake. This map serves as a surrogate for the functional integrity of the nigrostriatal dopaminergic projection (12).

The European Medicines Agency approved ^{123}I -ioflupane in 2000 (trade name DaTscan); the U.S. FDA followed in 2011 (trade name DaTscan). It remains the only approved in vivo imaging biomarker for parkinsonian syndromes in clinical practice.

2.2.2. Image Characteristics

In a healthy brain, DaTscan produces a characteristic pattern of high bilateral tracer uptake forming “comma” shapes: the dense head of each comma corresponds to the caudate nucleus, and the tapered body to the posterior putamen. In PD, because neurodegeneration begins predominantly in the posterior putamen (the subregion receiving projections from the motor cortex) the comma loses its tail before it loses its head, resulting in an asymmetric pattern progressively reduced to isolated caudate (12). In advanced disease, uptake is globally and bilaterally diminished. This topographic progression is the visual basis on which clinical readers classify scans.

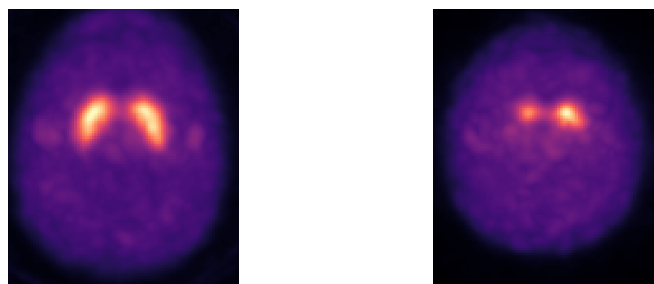


FIGURE 2: side-by-side DaTscan **coronal** slices showing normal bilateral *comma* pattern (left, healthy patient) vs. asymmetric *period* pattern of PD (right, diagnosed patient).

Conditions that spare the presynaptic dopaminergic terminal, notably essential tremor and drug-induced parkinsonism, produce normal DaTscan images, making the scan a powerful discriminator between dopamine-deficiency and non-deficiency parkinsonism, even when clinical features are indistinguishable.

2.2.3. Semi-Quantitative Analysis

The spatial complexity of DaTscan images is commonly reduced to a set of scalar metrics. The Striatal Binding Ratio (SBR) is the most widely used paradigm, defined as:

$$\text{SBR} = \frac{\text{Striatal counts} - \text{Background counts}}{\text{Background counts}},$$

where background activity is typically measured in the occipital cortex, a region demonstrating negligible DAT expression. SBR values are computed separately for the caudate, anterior putamen, and posterior putamen within each hemisphere, as implemented in the DaTQUANT software utilized by the PPMI dataset (13,14).

Semi-quantitative SBR provides an objective, reproducible numerical summary of tracer binding that correlates with disease severity and disease duration, and forms the basis of the classical machine learning baseline in this thesis. However, it compresses the full spatial information of the three-dimensional SPECT volume into at most a handful of regional means, inevitably discarding information about the spatial distribution of uptake within regions, asymmetry patterns, and subtle texture changes that may be diagnostically informative.

These SBR-derived features constitute the basis of the classical ML baselines evaluated in this thesis.

2.2.4. Clinical Utility and Limitations

A 2021 systematic review confirmed that DaTscan led to a change in clinical management in approximately half of patients tested and altered the final diagnosis in roughly one third (15), evidence of its practical diagnostic impact. Nevertheless, several limitations apply. DaTscan cannot differentiate PD from other neurodegenerative parkinsonian syndromes, all of which reduce nigrostriatal terminal density. Furthermore, like any imaging biomarker, DaTscan is normal in the presymptomatic phase before sufficient neurodegeneration has accumulated, and in rare cases may be normal even in clinically confirmed PD. These limitations motivate automated image-based approaches.

Hypothesis and Objectives

SECTION 3

3.1. Research Question

Current clinical interpretation of DaTscan images relies on subjective visual assessment by nuclear medicine specialists, supplemented in research settings by semi-quantitative metrics such as the Striatal Binding Ratio (SBR) computed from predefined regions of interest. The issue, as mentioned, is that traditional semi-quantitative metrics may discard diagnostically relevant spatial information. This thesis addresses the following research question:

“Can end-to-end CNNs operating on the complete DaTscan image volume perform better than traditional machine learning methods that use manually crafted semi-quantitative features for the binary classification of Parkinson’s disease compared to healthy controls? Additionally, does incorporating multimodal clinical variables offer a significant diagnostic advantage over imaging alone?”

3.2. Hypothesis

The central hypothesis is that a CNN trained end-to-end on DaTscan images will learn richer and more discriminative spatial representations than those captured by semi-quantitative SBR features, and that performance can be further improved by incorporating complementary clinical information that is not visible in the image. This is operationalized through the following specific hypotheses:

- H1** CNN superiority over classical baselines. A CNN classifier operating on the raw DaTscan volume will achieve higher AUC and classification accuracy than classical machine learning models (SVM, Logistic Regression) trained on SBR-derived features.
- H2** Transfer learning compensates for data scarcity. In a data-limited regime (~250 subjects), leveraging pretrained weights will improve generalization relative to training a 3D CNN from scratch, since pretrained representations provide a better initialization than random weights.
- H3** Raw images are more informative than registered images. DaTscan images in native acquisition space (unregistered) will yield equal or better classification performance than spatially normalized images.
- H4** Multimodal fusion improves upon imaging alone. Combining DaTscan-derived image embeddings with tabular clinical variables — motor function (MDS-UPDRS III), olfactory sensitivity (UPSIT), cognitive screening (MoCA), and demographics (age, sex) — will yield higher classification performance than either modality alone.

3.3. Objectives

3.3.1. General Objective

To systematically evaluate deep learning-based approaches for automated classification of Parkinson’s disease from DaTscan SPECT images, and to quantify the diagnostic information gain achievable through multimodal fusion with clinical variables, using the PPMI dataset as a standardized benchmark.

3.3.2. Specific Objectives

1. **Establish a classical ML baseline.** Train and evaluate SVM and Logistic Regression classifiers on semi-quantitative DaTscan features (SBR values) and multimodal clinical variables from the PPMI dataset.
2. **Compare CNN architectures of different dimensionality and pretraining strategy.** Implement and evaluate multiple CNN variants. Identify which combination of dimensionality and pretraining strategy best suits the available data size.
3. **Assess the effect of spatial registration.** Compare classification performance on registered (spatially normalized) versus raw (native space) DaTscan images to determine whether spatial normalization is beneficial or detrimental in this context.
4. **Evaluate multimodal fusion strategies.** Implement and compare fusion strategies to quantify the marginal diagnostic value of each clinical modality.
5. **Ensure ethical and reproducible use of data.** Conduct all experiments in accordance with the PPMI data use agreement, using stratified cross-validation to avoid data leakage,

3.4. Scientific Relevance and Expected Impact

The clinical motivation for this work is immediate. DaTscan is currently interpreted by visual assessment in most clinical centres, a process that is time-intensive, requires specialized expertise, and produces non-negligible inter-reader disagreement — particularly for borderline cases in early disease. An automated, objective classifier that operates directly on the full image volume could serve as a consistent second reader, reducing diagnostic uncertainty and standardizing reporting across sites. In the longer term, such tools could lower the expertise threshold required to interpret dopamine transporter scans, broadening their clinical utility.

From a methodological standpoint, this thesis contributes to the growing literature on CNN-based DaTscan analysis in several ways. The systematic comparison of 2D, 2.5D, and 3D approaches under controlled conditions — same dataset, same cross-validation scheme, same evaluation metrics — provides evidence that is directly actionable for practitioners designing similar systems. The finding that a 2.5D ImageNet-pretrained approach can outperform a purpose-built 3D medical network is counterintuitive and speaks to the critical role of dataset size in architecture selection: strong priors from large-scale pretraining may outweigh domain specificity when labeled medical data are scarce. The comparison of registered versus raw images similarly challenges a common preprocessing assumption in neuroimaging pipelines. And the systematic decomposition of multimodal gains by clinical feature group provides quantitative evidence for the relative diagnostic contribution of motor, olfactory, cognitive, and demographic variables — information that is clinically interpretable and actionable.

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Planning

APPENDIX A

The development timeline of the thesis is summarized in FIGURE 3 The proposal was submitted in late January and formally approved in early March. Preprocessing of the PPMI dataset and implementation of the classical machine learning baselines began shortly afterward, followed by development and training of the CNN architectures throughout March and April. Multimodal fusion experiments were conducted during early May. Thesis writing was carried out progressively based on implementation logs and experimental notes collected during the semester, while the Typst template and report structure were finalized in parallel.

Writing of this thesis was started in may based on log notes kept during the semester, while the template itself was finalised around april.

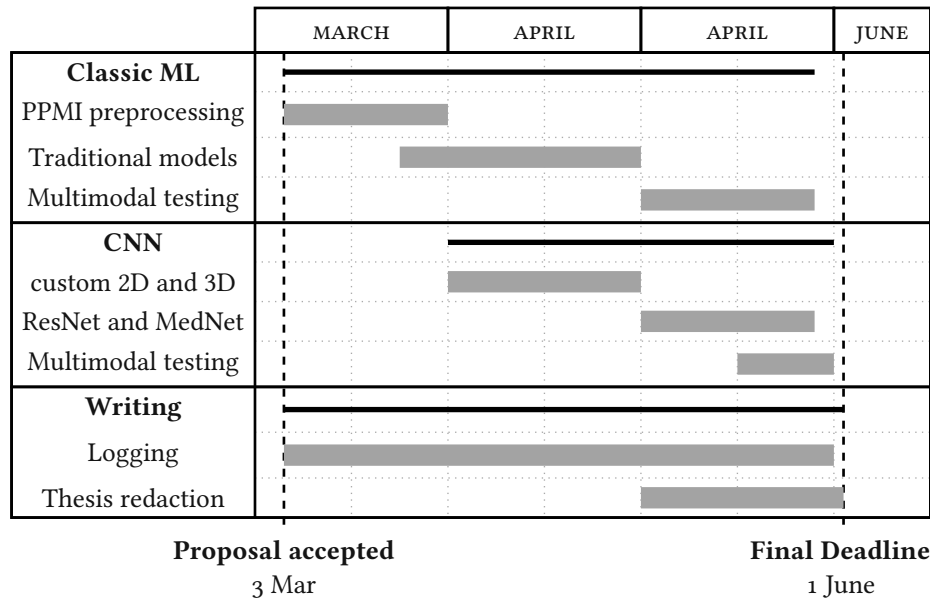


FIGURE 3: Gantt chart summarizing the main development stages of the thesis project.

Code

APPENDIX B

This project was developed under continuous version control using Git and is hosted on GitHub at [arnauKL/pd_datscan_cnn_thesis](https://github.com/arnauKL/pd_datscan_cnn_thesis). The repository contains the preprocessing pipeline, model implementations, training scripts, evaluation utilities, and experiment configuration files used throughout the thesis. The source code is distributed under the BSD 3-Clause License.